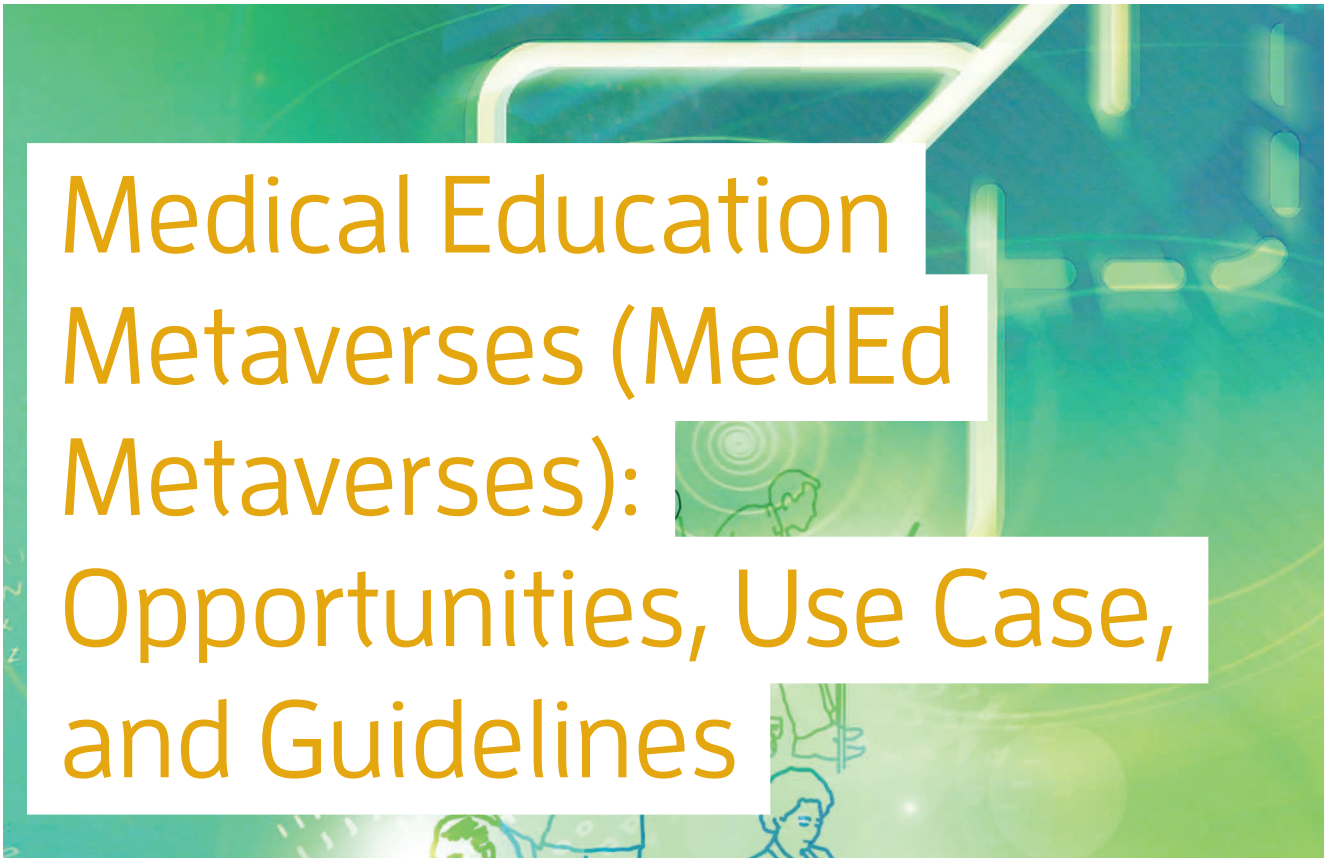


# Medical Education Metaverses (MedEd Metaverses): Opportunities, Use Case, and Guidelines

**Yosra Magdi Mekki**, Qatar University
**Leslie V. Simon** and **William D. Freeman**, Mayo Clinic
**Junaid Qadir**, Qatar University

*This article explores how artificial intelligence (AI), particularly generative AI (GenAI), can enhance extended reality (XR) applications in medical education (MedEd) metaverses. We compare traditional augmented reality/virtual reality methods with AI-enabled XR metaverses, highlighting improvements in immersive learning, adaptive feedback, personalized performance tracking, remote training, and resource efficiency.*

Despite the hype, the metaverse is not uniquely defined. Some pundits equate it to extended reality (XR), which itself is an umbrella term referring to augmented reality (AR), virtual reality (VR), and mixed reality (MR). Others equate it with "spatial computing," which combines physical spaces, digital computation, and virtual entities to create an immersive experience. This approach addresses the limitations of the Internet and mobile phones, which, while transformative, restrict us to small screens and detract from our natural experience of the spatial world.

Broadly speaking, the metaverse is widely accepted as a persistent, immersive media environment, enabling multiple users to experience it simultaneously with a strong sense of presence. It is envisioned as a 3D

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counterpart to the current 2D Internet, leveraging AI and XR technologies. Future metaverses will seamlessly blend our digital and physical lives, moving beyond the limitations of small screens. Generative AI (GenAI), using techniques like transformers, diffusion models, generative adversarial networks (GANs), and variational autoencoders, along with XR, will soon create entire virtual worlds from simple prompts or 2D images. Ng[1] provides a comprehensive conceptualization of the metaverse, which we adopt to clarify the technological framework of our study. Unlike Meta (formerly Facebook), the metaverse refers to a persistent, immersive virtual environment integrating AI, AR, VR, and MR, offering a broader and more accurate context for our discussion.

Graylin and Rosenberg, in their book *Our Next Reality*,[3] simplify this into a formula describing the metaverse as the 3D version of the global Internet we know, powered by AI and primarily accessed using XR devices as follows:

$$\text{Metaverse} = \text{Internet (3D)}^{\text{AI}} \leftrightarrow \text{XR}$$

This formula highlights the pivotal role AI and XR technologies will play in developing 3D immersive experiences central to the metaverse. XR technologies transform our world interaction by creating immersive experiences that augment or reimagine reality. VR offers a first-person, interactive simulated environment with a strong sense of presence in a digital world. AR places digital information or objects in our physical surroundings, while VR immerses users in entirely digital environments. The XR Association (XRA) defines MR as involving advanced systems where users interact with responsive virtual objects integrated into the real world. The reality to virtuality continuum illustrates the shift from physical spaces through AR and MR to fully virtual environments, as depicted in Figure 1. Advanced AI–XR technology offers unique capabilities, as shown in Table 1, enhancing AI–XR metaverses beyond traditional AR/VR and enabling new pedagogical opportunities, including adaptive learning, personalized feedback, and dynamic, learner-specific content in enriched immersive environments.

## MedEd Meets Metaverse

The Metaverse offers transformative potential for medical education by providing immersive, interactive learning environments. It enables lifelike simulations for various medical scenarios, allowing students to practice procedures and decision-making in a safe, controlled setting. These simulations replicate complex surgical interventions, emergency responses, and routine medical procedures, improving proficiency and confidence. AR and VR assist in surgical training with real-time guidance and visualization. The Metaverse supports high-fidelity manikins and virtual patients, offering detailed models for hands-on practice. Personalized learning interventions tailored to individual needs enhance the educational experience through targeted feedback and adaptive learning paths. For continuing medical education, the Metaverse offers virtual workshops, conferences, and collaborative spaces for knowledge sharing and staying updated on advancements. Incorporating the Metaverse into MedEd ensures future health-care professionals are better prepared, more skilled, and equipped with the latest technological knowledge to improve patient outcomes and safety. However, challenges remain as noted by Curran and Hollett[4] argue that XR technologies, despite their potential for engaging patient education, face challenges in widespread adoption, particularly among marginalized populations.

This article contributes to the discussion by exploring the transformative role of the Metaverse in medical education. It provides a comprehensive review of both the opportunities and risks presented by MedEd Metaverses, offering insights into their potential benefits as well as limitations. We also present a detailed use case that demonstrates practical applications and benefits of AI–XR MedEd Metaverses, illustrating how they can create flexible, immersive learning experiences.

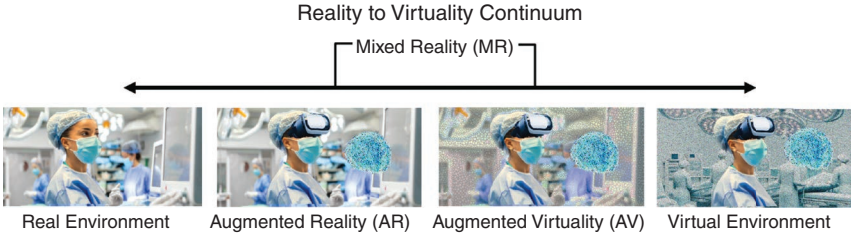



**FIGURE 1.** Illustration of the reality to virtuality continuum in surgery, showing a trainee using headsets and modules to gain intraoperative insights. The continuum depicts the shift from the physical operating room (OR) to a virtual OR. (Adaptation from the Milgram and Kishino (1994) reality to virtuality continuum.[2])

Additionally, the article provides a set of guidelines and best practices for effectively integrating Metaverse technologies into medical education curricula, aimed at supporting instructors, students, and developers in navigating this emerging landscape.

The article begins with an overview of MedEd Metaverses and AI-enabled MedEd Metaverses (The "MedEd Metaverses: Background" section, followed by an analysis of the opportunities and risks associated with these technologies in the "MedEd Metaverses: Opportunities And Risks" Section. A use case showcasing the use of AI–XR MedEd Metaverse for creating a flexible learning experience is described in the "Use Case: Showcasing an AI-XR MedEd Metaverse Platform" section. Guidelines and best practices for instructors, students as well as MedEd metaverse developers are discussed in the "MedEd Metaverses: Tips and Guidelines" section. Finally, the article is concluded in the last section.

## MedEd METAVERSES: BACKGROUND

The metaverse offers transformative possibilities for medical education by providing immersive, interactive, and highly effective training environments. As these technologies evolve, their role in medical training and health-care delivery will likely expand, offering significant benefits for students and professionals. AR and VR technologies have become integral tools in medical education, providing immersive and interactive learning experiences that enhance traditional teaching methods, including improved understanding, skill enhancement, and better patient outcomes. These advanced technologies are applied across various domains, enhancing medical students' and professionals' learning processes and skill acquisition.

### Existing commercial MedEd AR/VR applications

While numerous AR/VR applications exist in medical education (MedEd), the scope of this article is not to exhaustively survey them but to highlight the new pedagogical possibilities enabled by AI–XR MedEd metaverses, supported by selected salient references.

**TABLE 1. Comparative analysis of learning experiences for medical trainees using traditional AR/VR versus AI-enhanced XR metaverses in medical education.**

| Feature | Traditional AR/VR | AI-Enhanced XR Metaverse |
|---|---|---|
| Immersive learning experiences | Simulates predefined medical procedures with limited adaptability | Uses AI to adapt simulations based on learner interactions, enhancing immersion through context-aware scenarios and personalized feedback |
| Suspension of disbelief | Combines real and virtual elements to create consistent, static learning environments | Integrates AI for dynamic adjustments of virtual and real elements, enhancing engagement by tailoring content to the user's responses and actions |
| Haptic feedback | Provides basic haptic feedback for simulating texture and resistance | Employs AI-driven haptic adjustments that adapt in real-time, providing more realistic and context-sensitive tactile interactions, tailored to the learner's actions |
| Performance tracking | Allows tracking of student progress using general metrics like speed and accuracy | Uses AI to provide in-depth analysis of learner performance, offering adaptive, real-time feedback and targeted exercises for improved learning outcomes |
| Remote training | Offers remote access to standardized medical training modules | AI optimizes remote training by adjusting module difficulty, providing personalized content, and offering intelligent tutoring based on real-time learner data |
| Resource efficiency | Reduces the need for physical resources by providing digital simulations. | Enhances efficiency through intelligent resource allocation—adapting training sessions to reduce redundancy and optimize individual learning time, saving both time and costs. |

Currently, most of the leading MedEd AR/VR applications that enhance the learning experience by offering immersive, interactive environments are commercial. This landscape will be reviewed next.

**Enhanced anatomy education.** VR applications like 3D Organon allow students to explore detailed 3D anatomical models, enhancing understanding beyond traditional cadaver dissection. AR applications like HoloAnatomy overlay digital anatomical models onto the physical world, facilitating studies using smartphones or tablets. Taylor et al.[5] have demonstrated the effectiveness of AR as an independent study tool and highlighted the popularity of VR as a supplement for teaching anatomy.

**Improved surgical training.** VR platforms like OssoVR provide immersive surgical simulations, allowing trainees to practice procedures in a risk-free environment. AR applications like Medivis overlay digital information onto the surgeon's view, offering real-time guidance and enhancing precision.

**Clinical skills training enhancement.** VR patient simulators like Lumeto[6] offer virtual patient interactions, allowing students to practice clinical skills, communication, and decision-making in a controlled environment. As described by Herur-Raman et al.,[7] the integration of XR into medical education can enhance empathy and communication skills. AR tools like Proximie overlay diagnostic information during assessments, enhancing practical learning. Effective communication is crucial in health care to minimize patient risk and avoid miscommunication.

**Emergency response training.** In emergency response training, VR platforms like Lumeto and Oxford Medical Simulations simulate emergency scenarios, such as natural disasters or mass casualty incidents. These allow health-care students and professionals to practice response strategies and teamwork in a safe and controlled environment. AR applications such as Proximie can simulate triage situations, helping trainees practice and improve emergency decision-making skills.

**Patient education.** Patient education is another area where these technologies prove valuable. VR can educate patients about their conditions and treatment options; for instance, Aspetar's VR Patient Education Clinic can simulate the effects of diseases and demonstrate surgical procedures. AR applications such as EchoPixel and AccuVein provide patients with interactive information about their health and treatments, enhancing understanding and compliance.

## AI-enabled MedEd Metaverses

AI-enabled educational metaverses are set to transform medical education, particularly due to the rapid advancements in GenAI. Beyond simply utilizing AR/VR platforms to deliver medical education, an AI-enabled metaverse represents a significant leap forward in providing tools and capabilities that were previously unimaginable, bringing a more personalized learning experience for future medical professionals.

**Enhanced realism and interactivity.** GenAI can dynamically create realistic and complex virtual medical environments, offering richer, procedurally generated simulation scenarios tailored to individual students' needs.

> **THE METAVERSE OFFERS TRANSFORMATIVE POSSIBILITIES FOR MEDICAL EDUCATION BY PROVIDING IMMERSIVE, INTERACTIVE, AND HIGHLY EFFECTIVE TRAINING ENVIRONMENTS.**

This includes varied medical scenarios, including rare or complex cases, ensuring students are exposed to various conditions. Immediate, detailed feedback on student performance can be provided by AI during simulations, analyzing techniques in real time, pointing out errors, and suggesting improvements, thus enhancing the learning experience.[8]

**Intelligent virtual patients.** Virtual patients in medical training can be significantly enhanced through large language models (LLMs), making them lifelike and capable of meaningful interactions. These AI-driven patients can dynamically adapt symptoms and responses based on student interactions, enhancing immersive

learning. Feedback-enabled LLMs simulate realistic patient behaviors, allowing the practice of technical skills and essential soft skills like communication, empathy, and decision-making. The "Virtual You" concept[9] uses digital twins[10] for personalized, predictive, and preventive health-care solutions. Healthcasts, real-time broadcasts of health status from digital twins, enable continuous monitoring and insights into health trajectories. This allows counterfactual analysis by simulating various health scenarios, helping students explore "what-if" situations, such as lifestyle changes on chronic diseases or treatment plan outcomes, offering valuable learning experiences.

**Advanced haptic feedback and sensory experiences.** AI enhances haptic feedback to provide realistic tactile sensations, making virtual medical procedures feel more lifelike. By integrating visual, auditory, and haptic feedback, AI creates immersive multisensory experiences, crucial for preparing students for real-life medical situations and reinforcing practical skills. Bidirectional interfacing of AR/VR with spatial technologies enables precise control of humanoid robotics, supported by high-speed networks and satellite Internet for seamless interaction.[11,12,13,14] This integration allows for effective remote operations with real-time feedback.

**Better integration of real and virtual worlds.** Advancements in AI are enabling closer integration of physical and virtual worlds, enhancing mixed-reality experiences in applications like remote surgery, virtual meetings, and collaborative workspaces. Progress in photorealistic avatars, generative AI, and LLMs has improved the realism and interactivity of virtual elements. AI-driven spatial computing adds context-awareness to these components, making them more effective for medical training. Despite these advances, full seamless integration remains a challenge at the current stage of development.

**Content creation and personalization.** GenAI can generate a wide range of medical content, from virtual anatomical models to interactive case studies, reducing the need for manual creation and enabling dynamic, diverse learning experiences. AI tools empower students and educators to easily create virtual medical training experiences and assets, fostering a participatory and creative learning environment. This capability ensures educational content is always up-to-date and relevant. By analyzing a student's progress, the model can adjust the difficulty and type of content, optimizing the learning path and helping students grasp complex concepts more effectively.

**Global collaborative networks.** The AI-enabled metaverse can facilitate global collaboration among medical students, educators, and professionals. Virtual classrooms and labs can connect individuals worldwide, allowing for the exchange of knowledge, best practices, and cultural perspectives in medicine. Students can work together on interactive case studies in a virtual environment, simulating real-world medical teamwork. A GenAI-based proctor can moderate these sessions, providing guidance and meeting learning objectives. In a traditional medical multidisciplinary meeting, various specialists from different fields come together to discuss complex cases, particularly in the context of interprofessional education scenarios.

**Swarm intelligence in medical collaboration.** Swarm intelligence involves the collective behavior of decentralized, self-organized systems, like ant colonies or bird flocks. It can enhance decision-making and problem-solving by combining human expertise with AI-powered avatars in shared spaces. AI avatars, programmed with specialized medical knowledge, can join multidisciplinary meetings, offering data-driven insights, predictive analytics, and real-time information processing. They interact seamlessly with human practitioners, sharing information, simulating complex scenarios, providing instant feedback, and suggesting alternative treatment plans. The scalability of AI avatars allows them to be replicated and tailored for different meetings.

## MedEd Metaverses: Opportunities and Risks

### Benefits and opportunities

AI–XR metaverse-enabled education offers numerous benefits for medical students and professionals as showcased in Figure 2. These technologies create immersive, interactive learning environments, allowing deep engagement with complex medical concepts and procedures. They facilitate global collaboration and remote learning, breaking geographical barriers and providing high-quality education to underserved areas. AI customizes educational content to individual learning styles, offering personalized feedback and adaptive learning paths. Simulating high-risk medical scenarios in a controlled environment helps students develop critical skills and confidence without associated risks. MedEd Metaverses provide realistic simulations for practicing medical

procedures, from basic anatomy to complex surgeries, blending physical and digital worlds effectively. Additionally, real-time data analytics continuously assess student performance, providing timely feedback and identifying areas for improvement.

## Risks and challenges

While AI–XR technologies offer numerous benefits in medical education, they also present significant risks and challenges.[15] Overreliance on AI for diagnostics and decision-making can diminish hands-on experience and critical thinking skills among medical professionals. Extensive data collection for AI-driven personalization and assessment raises privacy and ethical concerns, including potential breaches of sensitive information.[16] Compliance with General Data Protection Regulation (GDPR) and the Health Insurance Portability and Accountability Act (HIPAA) is crucial, as AI–XR metaverses involve intimate surveillance, heightening privacy risks.[17] Adhering to these regulations is challenging but vital for safeguarding user data in health care. Disparities in access to technology and high-speed Internet exacerbate educational inequalities, disadvantaging students in underprivileged areas. Additionally, prolonged use of virtual environments may lead to social isolation, addiction, and other mental health issues. Addressing these challenges requires a balanced blend of technical advancements and regulatory measures.[18] Major corporations such as Apple, Meta, Microsoft, and others drive technological advancement in the metaverse through substantial resources and expertise, creating economic opportunities and new job markets. However, their profit motives may overshadow educational benefits, influencing user behavior. In particular, corporate control of metaverse technologies presents risks like fine-grained tracking and behavior manipulation, necessitating appropriate regulation to ensure prosocial outcomes and prevent misuse.[3]

To mitigate potential risks, AI–XR MedEd designers should consider the following questions—developed based on our analysis and experience as practitioners and consultants in this domain—before deployment to ensure metaverses in medical education enhance learning while safeguarding ethical and practical concerns:

1. Does the module enhance engagement with the physical world and real-world healthcare issues?
2. Will the MedEd Metaverse amplify existing real-world inequality?
3. Have the system-level repercussions of XR technology usage been considered?
4. What policy and regulatory frameworks have been considered to ensure no violation of human rights?
5. Have the necessary steps been taken to accommodate and promote diversity and inclusion so that the developed technology is appropriate for the needs and sensitivities of different groups?
6. Is there transparency in the system regarding what data is collected, who it is shared with, and how it is used?
7. Is the technology culturally appropriate?

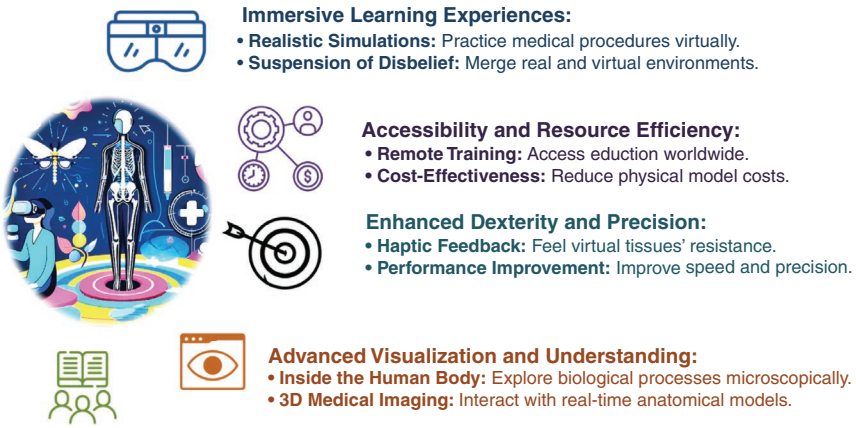



**FIGURE 2.** Benefits of AI–XR MedEd Metaverses for medical education, spanning immersion, visualization, and understanding.

## USE CASE: SHOWCASING AN AI–XR MEDED METAVERSE PLATFORM

The purpose of this use case is to concretize the potential of the AI–XR MedEd metaverse by illustrating a practical application scenario. We discuss how modern MedEd metaverse capabilities build upon traditional medical simulation, offering expanded possibilities such as immersive learning and interactive, flexible education. This section also presents an AI-enabled

MedEd metaverse implementation, featuring customizable and structured learning experiences designed to cater to diverse learner needs. Our aim is illustrative, providing insight into how these technologies can transform medical education, with plans for a more systematic evaluation of the implementation in future work.

## Advancing medical simulation with MedEd Metaverses

Historically, medical simulation has relied on manikins and simulated patients (SPs). Manikin-based training lack in monitoring and assessing nonverbal communication, while SPs, despite offering live interactions, pose logistical and financial challenges. Conventional XR-based solutions often use prescripted, uneditable dialog prompts, limiting their effectiveness in developing communication skills. AI-enabled metaverses, combining GenAI and XR, facilitate natural, interactive communication between health-care learners and virtual patients. This capability enhances competency in patient engagement and care, creating immersive learning environments that foster psychological safety. By simulating real-world care nuances, these platforms support the development of critical competencies including procedural knowledge, clinical decision-making, communication effectiveness, and emergency response, ultimately aiming to enhance patient safety and health-care quality.

## AI-enabled MedEd Metaverse implementation

This use case highlights how AI-enabled metaverses can transform medical education, enhancing the preparedness and confidence of health-care professionals while ensuring high-quality patient care. It explores how an AI-enabled VR metaverse platform uses GenAI to address these limitations innovatively. This use case demonstrates the potential of our AI–XR system, highlighting its innovative application, while empirical validation will be addressed through future systematic evaluations.

One example of an AI-enabled metaverse in medical education is Lumeto's InvolveXR, which integrates an LLM-based system for highly realistic and interactive patient simulations. This technology supports natural language processing, allowing virtual patients to respond in a human-like manner, enhancing the authenticity and effectiveness of the training experience. It also reduces the cognitive load on educators, freeing them from simulating patient interactions or checking dialog prompts, allowing more focus on learners and their outcomes.

At our medical school workshops, we have used in-house and commercial applications to create simulation experiences for teaching and assessing trainees in tasks like history-taking and physical examinations. We employed Lumeto's world-building tools for rapid development, meeting instructors' specific needs efficiently. Our use case featured immersive scenarios mirroring real medical situations, including customized patient avatars, detailed medical histories, and interactive examination routines that enhance engagement and education.

In the MedEd Metaverse, learners engage with realistic, interactive simulations that replicate real-life medical scenarios. They practice virtual physical exams and history-taking, receiving instant feedback on their decisions. This approach enhances critical thinking and clinical reasoning skills by requiring active participation in diagnostic processes and interventions. Instructors use Lumeto's tools to create customized scenarios, monitor learner progress in real time, and provide personalized guidance. Detailed analytics track task completion and diagnostic accuracy, while collaborative scenarios simulate multidisciplinary team interactions, improving communication and decision-making in clinical settings.

By empowering instructors to create and modify simulations directly, we tackled the challenge of aligning expectations between medical professionals and XR developers. Our work showcased that intuitive world-building tools can swiftly and effectively produce high-quality medical simulations, enriching trainee learning experiences. Looking ahead, we advocate for simulation and XR labs in medical schools to prioritize intuitive, ideally open source, tool development. This fosters collaboration and innovation in medical education while reducing dependence on proprietary software and associated risks.

Figure 3 shows a learner's immersive experience in a digital metaverse for medical education. The AI–XR platform facilitates realistic, interactive scenarios, where spoken input is converted to text, and the patient responds dynamically. The learner receives contextual information and prompts to improve diagnostic skills. Detailed feedback and supplementary materials support decision-making, highlighting the effectiveness of AI–XR technology in medical training.

## Customized flexible structured learning experiences

AI-enabled XR metaverses allow educators to adapt scenarios for various educational levels, from nursing schools to fellowship programs. These platforms

support both synchronous and asynchronous learning, offering flexible training for in-person and remote learners. By creating diverse patient profiles, including history, lifestyle, and accents, they enhance cultural competency and inclusion. This adaptability helps learners practice critical skills like clinical decision-making, communication, and emergency response in a safe environment. Platforms like Lumeto's InvolveXR use frameworks like Bloom's Taxonomy to structure learning, guiding students from foundational knowledge to advanced problem-solving. This approach fosters critical thinking and key professional skills. As shown in Figure 4, instructors can track learners' progress using such tools in real time using an event log and assessment checklist, which makes the platform more effective and encourages frequent use for simulations. Such platforms can also effectively support multidisciplinary collaboration, which is fundamental in real-world health-care settings. Figure 5 demonstrates how a multidisciplinary approach can be integrated into virtual medical training. By fostering collaboration across specialties, AI–XR metaverses prepare learners for team-based patient care, enhancing their educational experience and readiness for comprehensive management.

## MedEd Metaverses: Tips and Guidelines

### MedEd Metaverses: Instructor guidelines and best practices

These guidelines are based on the authors' experience in AR/VR/XR and medical education, supported by a review of relevant literature. They aim to help educators and students effectively use AI–XR technologies in

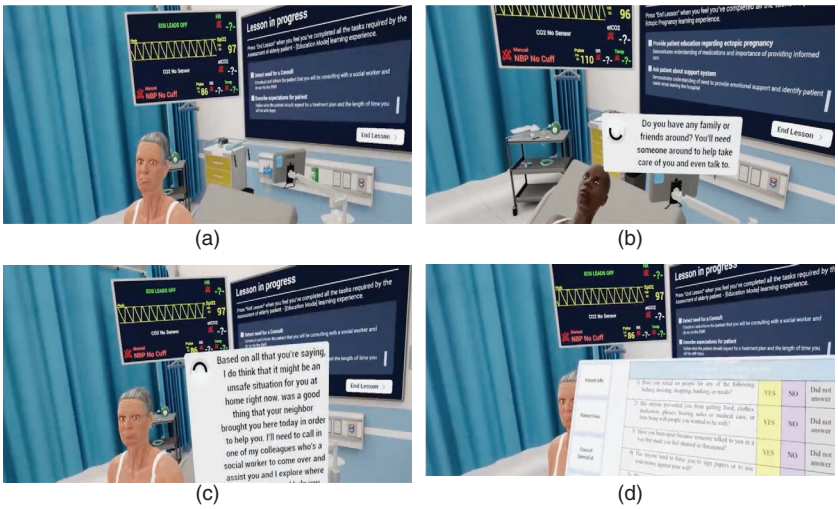



**FIGURE 3.** Learner viewpoint from headset. The learner can speak to the patient, whereby speech is converted to text and displayed, and the patient replies. The learner can also view complimentary material to aid in asking the right questions, illustrating the feasibility of a MedEd AI–XR metaverse. (a) Learner interacting with GenAI–enabled virtual patient. (b) Patient responds dynamically. (c) Display of speech-to-text for learner questions. (d) Supplementary materials displayed where needed for learner's convenience.

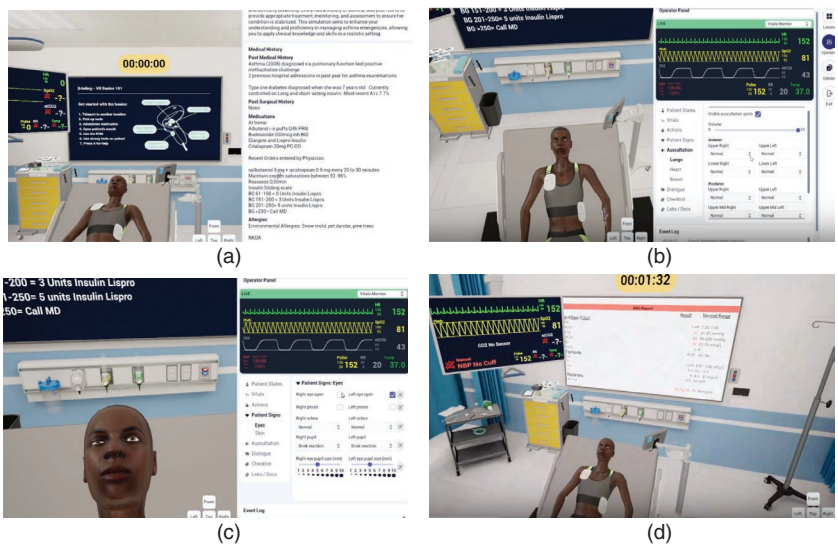


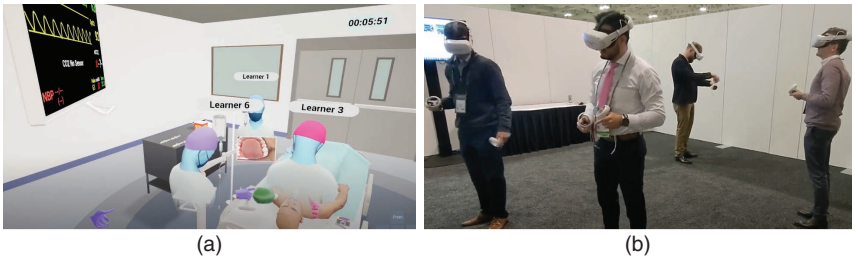

**FIGURE 4.** Instructor viewpoint from headset. It is important to note that simulation instructors value intuitive platforms with instantaneous tracking and customization of the training simulation scenario. (a) Instructor controls information engagement with patient. (b) The ability to modify parameters such as lung sounds during the simulation is a positive adoption factor for instructors. (c) Modifying patient signs and tracking learner performance accordingly in an event log. (d) Monitoring and assessment of the simulation on a PC device.

MedEd metaverses and contribute to discussions on improving pedagogical practices in medical education.

1. *Adaptive Teaching Methods*: Adapt your teaching style to the needs of your students. Some may prefer visual aids, while others benefit from hands-on practice. Utilize XR's versatility to cater to different learning preferences.
2. *Engagement and Motivation*: Keep students engaged by making scenarios interactive and challenging. Use gamification elements like rewards or progress tracking to motivate learners.
3. *Cultural Sensitivity*: Be mindful of cultural differences and sensitivities when designing and conducting simulations. Ensure that scenarios are inclusive and respectful of diverse backgrounds.
4. *Safety and Psychological Comfort*: Create a safe learning environment where students feel comfortable making mistakes and learning from them. Address any anxiety related to performance or the use of new technology.
5. *Feedback Mechanisms*: Use various feedback methods, including self-assessment, peer feedback, and instructor evaluations. Incorporate technology to provide instant feedback during simulations.
6. *Professional Development*: Continually update your knowledge and skills in medical practice and XR technology. Attend workshops, conferences, and training sessions to stay current with advancements in both fields.
7. *Spotlight Approach*: Try to use "world-building" platforms to customize the experience to exactly what you have in mind, rather than relying on out-of-the-box experiences. Mekki has written a piece that sheds light on the importance of focusing on such platforms in AI.[19]

## MedEd Metaverses: Student guidelines and best practices

1. *Active Participation*: Engage actively in simulations. The more you immerse yourself in the scenario, the more you will benefit from the experience.
2. *Continuous Learning*: Use XR simulations to complement traditional study methods. Review related materials before and after simulations to reinforce learning.
3. *Peer Collaboration*: Work closely with your peers during simulations. Share insights, ask questions, and learn from each other's experiences.
4. *Self-Assessment*: After each simulation, assess your performance critically. Identify areas where you excelled and where you need improvement.
5. *Technology Proficiency*: Take the time to become proficient with the XR equipment and software. Familiarity with the tools will help you focus on learning rather than troubleshooting technical issues.
6. *Stress Management*: Learn and practice stress management techniques. Breathing exercises, mindfulness, and relaxation techniques can help you stay calm and focused during high-pressure simulations.
7. *Experiment*: The current availability of more user-friendly tools makes this an opportune time to develop XR applications, as evidenced by the success of medical students in building models.[20] By mastering game engines such as Unity, one can create educational experiences that surpass the constraints of standard world-building tools, offering greater flexibility and creativity.




(a) (b)

**FIGURE 5.** Multidisciplinary approach to patient care within MedEd Metaverses. (a) Social presence and relative position tracking. (b) Medical professionals using Lumeto for training.

## MedEd Metaverse: Developer guidelines and best practices

1. *User-Centric Design*: Design simulations with the end-user

in mind. Gather feedback from both instructors and students to understand their needs and pain points.

2. *Scalability and Flexibility*: Ensure that your XR platform is scalable and flexible. It should be able to accommodate a range of scenarios and be easily updated or modified.
3. *Interactivity and Realism*: Focus on creating highly interactive and realistic scenarios. Use high-quality graphics, accurate medical data, and realistic patient responses to enhance the learning experience.
4. *Performance Optimization*: Optimize the performance of your XR applications to ensure smooth and seamless user experiences. Address issues like latency, frame rate, and user interface responsiveness.
5. *Data Security and Privacy*: Implement robust data security measures to protect sensitive medical information and user data. Ensure compliance with relevant regulations and standards.
6. *Usability Testing*: Conduct extensive usability testing with actual users. Observe how they interact with the simulations and make iterative improvements based on their feedback.
7. *Educational Alignment*: Align your simulations with educational standards and curricula. Ensure that your content meets the learning objectives and competencies required by medical education programs.

## ABOUT THE AUTHORS


**YOSRA MAGDI MEKKI** is a final-year medical student at the College of Medicine, Qatar University, Doha, Qatar. Her research interest include developing and deploying machine learning and extended reality solutions for health. Contact her at ym1707134@qu.edu.qa.

**LESLIE V. SIMON** is an emergency medicine physician, associate professor, chair of emergency medicine, and director of the Simulation Center at the Mayo Clinic, Jacksonville, FL 32224 USA. Her research interests include medical simulation, difficult airway management, and patient safety. Simon received her D.O. from the Nova Southeastern University. Contact her at simon.leslie@mayo.edu.

**WILLIAM D. FREEMAN** is a professor of neurology and neurosurgery at the Mayo Clinic, Jacksonville, FL 32224 USA. His research interests include simulation, metaverse, and computational medicine. Freeman received his M.D. from the University of Texas at San Antonio. Contact him at freeman.william1@mayo.edu.

**JUNAID QADIR** is a professor of computer engineering at Qatar University, Doha, Qatar. His research interests include applied artificial intelligence (AI), AI ethics, and medical AI. Qadir received a Ph.D. in computer science and engineering from the University of New South Wales. He is a Senior Member of IEEE and ACM. Contact him at jqadir@qu.edu.qa.


In this article, we have explored how integrating GenAI and XR into metaverses can facilitate the development of MedEd Metaverses that can provide dynamic, personalized learning environments that adapt to students' needs and significantly enhance engagement. By adopting digital twins, virtual patients, and XR-enabled collaboration, educators can create more engaging and effective learning experiences. This article has highlighted both the opportunities and risks associated with these technologies, providing guidelines and best practices for instructors, learners, and AI–XR developers. Embracing these advancements is crucial for addressing current and future challenges in medical education, ensuring that students benefit from cutting-edge, immersive educational experiences. Future work will empirically evaluate the AI–XR system by focusing on student performance, engagement, and comparisons with existing XR technologies to assess its effectiveness. Moreover, advancements such as improved AI algorithms, enhanced VR experiences, and real-time integration from wearable devices aim to refine digital

twins, enabling more personalized care, better patient outcomes, and greater health-care efficiency.

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## REFERENCES

1. D. T. K. Ng, "What is the metaverse? definitions, technologies and the community of inquiry," *Australas. J. Educational Technol.*, vol. 38, no. 4, pp. 190–205, Jul. 2022, doi: 10.14742/ajet.7945.
2. P. Milgram and F. Kishino, "A taxonomy of mixed reality visual displays," *IEICE Trans. Inf. Syst.*, vol. 77, no. 12, pp. 1321–1329, Dec. 1994.
3. L. Graylin and L. Rosenberg, *Our Next Reality*. Boston, MA, USA: Nicholas Brealey, 2024.
4. V. R. Curran and A. Hollett, "The use of extended reality (XR) in patient education: A critical perspective," *Health Educ. J.*, vol. 83, no. 4, pp. 338–351, 2024, doi: 10.1177/00178969231198955.
5. L. Taylor, T. Dyer, M. Al-Azzawi, C. Smith, O. Nzeako, and Z. Shah, "Extended reality anatomy undergraduate teaching: A literature review on an alternative method of learning," *Ann. Anatomy*, vol. 239, Art no. 151817, Jan. 2022, doi: 10.1016/j.aanat.2021.151817.
6. T. Gore, "Lumeto Launches the First Customizable AI-Driven Conversational System in VR Healthcare Simulation." HealthySimulation. com. 2024. Accessed: Jun. 13, 2024. [Online]. https://www.healthysimulation.com/53116/lumeto-clinical-vr-ai-conversational-training/
7. A. Herur-Raman, N. D. Almeida, W. Greenleaf, D. Williams, A. Karshenas, and J. H. Sherman, "Next-generation simulation—integrating extended reality technology into medical education," *Frontiers Virtual Reality*, vol. 2, Art no. 693399, Sep. 2021, doi: 10.3389/frvir.2021.693399.
8. A. Qayyum et al., "Can we revitalize interventional healthcare with AI-XR surgical metaverses?," in *Proc IEEE Int. Conf. Metaverse Comput. Netw. Appl. (MetaCom)*, Kyoto, Japan, 2023, pp. 496–503, doi: 10.1109/MetaCom57706.2023.00091.
9. P. Coveney, R. Highfield, and V. Ramakrishnan, *Virtual You: How Building Your Digital Twin Will Revolutionize Medicine and Change Your Life*. Piscataway, NJ, USA: IEEE, 2023.
10. E. Katsoulakis et al., "Digital twins for health: A scoping review," *NPJ Digital Medicine*, vol. 7, no. 1, Art no. 77, Mar. 2024, doi: 10.1038/s41746-024-01073-0
11. "Humanplus: Humanoid shadowing and imitation from humans." GitHub, 2024. Accessed: Jun. 14, 2024. [Online]. Available: https://humanoid-ai.github.io/.
12. F. Sanfilippo et al., "A perspective review on integrating VR/AR with haptics into STEM education for multi-sensory learning," *Robotics*, vol. 11, no. 2, p. 41, 2022, doi: 10.3390/robotics11020041
13. D. Roberts and N. J. Roberts, "Maximising sensory learning through immersive education," *J. Nursing Educ. Practice*, vol. 4, no. 10, pp. 74–79, 2014, doi: 10.5430/jnep.v4n10p74
14. Y. Lin et al., "Satellite-MEC integration for 6G internet of things: Minimal structures, advances, and prospects," 2023, *arXiv Preprint arXiv:2308.14992*.
15. A. Tabassum and J. Qadir, "A critical AI approach for managing the risks in educational metaverses," in *Proc. 16th Int. Conf. Educ. New Learn. Technologies*, Palma, Spain, 2024, pp. 9578–9586, doi: 10.21125/edulearn.2024.2318
16. M. Alkaeed, A. Qayyum, and J. Qadir, "Privacy preservation in artificial intelligence and extended reality (AI-XR) metaverses: A survey," Sept. 2023, *arXiv:2310.10665*.
17. L. Rosenberg, "The metaverse and conversational AI as a threat vector for targeted influence," in *Proc. IEEE 13th Annu. Comput. Commun. Workshop Conf. (CCWC)*, Las Vegas, NV, USA, 2023, pp. 0504–0510, doi: 10.1109/CCWC57344.2023.10099167.
18. A. Tabassum, E. Elmahjub, and J. Qadir, "Pathway to prosocial AI-XR metaverses: A synergy of technical and regulatory approaches," in *Proc. Int. Symp. Netw. Comput. Commun. (ISNCC)*, Doha, Qatar, Oct. 2023, pp. 1–8, doi: 10.1109/ISNCC58260.2023.10323912.
19. Y. M. Mekki, "Physicians should build their own machine-learning models," *Patterns*, vol. 5, no. 3, Art no. 100948, Mar. 2024, doi: 10.1016/j.patter.2024.100948.
20. Y. M. Mekki, M. M. Mekki, M. A. Hammami, and S. M. Zughaier, "Virtual reality module depicting catheter-associated urinary tract infection as educational tool to reduce antibiotic resistant hospital-acquired bacterial infections," in *Proc. IEEE Int. Conf. Inform. IoT Enabling Technologies (ICIoT)*, Feb. 2020, pp. 544–548, doi: 10.1109/ICIoT48696.2020.9089488.